

# Performance audit report (sample deliverable)

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**Client:** Larkspur Pay, a fictional payments company. **System:** a credit-transfer pipeline of 4 Spring Boot services, Kafka and Postgres.

This is a sample deliverable produced from the lab in this repository. The client, the system and all data are fictional. Every figure below is generated from the committed benchmark results ( `results/` ) at git revisions `06ce6f4` , `8862be4` , `958b37b` . The report shows the method end to end: baseline, measure, diagnose, fix, re-measure, report. Read section 9 (limits of this evidence) before quoting any number.

## 1. Executive summary

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**Target.** This is a sample, so there is no client-specific target. The target used throughout is the objective in section 2 (end-to-end p99 under 500 ms, HTTP error rate under 0.1%, no dropped iterations) at the offered rates of the steady scenario.

**Capacity as found.** Under the stepped load, the baseline pipeline met the objective at every offered rate up to 200 req/s and missed it at 400 req/s. The true limit lies between those steps.

**After tuning.** With the 8 changes (F-01, F-03, F-04, F-06, F-07, F-08, F-12, F-13) applied, the tuned pipeline met the objective at every offered rate up to 800 req/s and missed it at 1200 req/s. The true limit lies between those steps. The step that missed it was offered only in the additional run `2026-09-30_tuned_steady_extended` . The two step ranges do not overlap, so the tuned profile sustains a clearly higher load, but the step lists are coarse, so no single "times faster" figure is given.

**Spike behaviour.** The burst raised the offered rate from 50 to 500 req/s for 60 s. Baseline result: not recovered within the observed window (the recovery phase lasted 120 s); the true recovery time is unknown. Tuned result: no degradation was observed, and payments created after the burst all met the SLO. The burst rate is below the highest step the tuned profile sustained in the steady runs, so the tuned system was never overloaded by the burst: this shows no degradation under the same load, not a faster recovery.

**Correctness.** Across all 10 recorded runs (729,781 payments created in total), no balance went negative, debits equalled credits, and every payment reached a terminal state (sections 4 and 8).

**Top 3 fixes** by assessed impact, then effort, then risk: F-04, F-06, F-03 (needs F-12, F-07 first). Ranked by impact against effort (section 7): quick wins F-03, F-06; major projects F-04, F-05 (not benchmarked), F-07, F-11 (not benchmarked), F-12. The recommended order differs from the ranking because of dependencies (for example, more consumer threads only help after more partitions and larger pools); section 7 gives the order.

**Expected gain.** Only the combined effect was measured (the step ranges above). The gain from each change on its own was not isolated, so no per-fix gain is claimed.

**What this report cannot tell you.** The combination of the 8 changes (F-01, F-03, F-04, F-06, F-07, F-08, F-12, F-13) in the recorded tuned runs was measured. The effect of each change alone was not isolated. No per-stage measurements (consumer lag, CPU, garbage collection, lock waits) were captured for these runs, so no single component is named as the bottleneck.

## 2. Scope and method

**Environment.** Apple M5, 10 cores, 16 GiB of memory, Darwin 27.0.0 arm64, Java 21.0.11, k6 v2.3.0. Postgres, Kafka, Redis, Prometheus, Grafana and the exporters run in containers with fixed limits; the services and the load generator run directly on the same machine, uncapped. Services and k6 run on the host and share CPU with the Docker VM; results are relative to this machine.

| Container         | CPU limit | Memory limit |
|-------------------|-----------|--------------|
| grafana           | 0.50      | 512 MiB      |
| kafka             | 2.00      | 2048 MiB     |
| kafka-exporter    | 0.25      | 128 MiB      |
| postgres          | 2.00      | 2048 MiB     |
| postgres-exporter | 0.25      | 128 MiB      |
| prometheus        | 0.50      | 512 MiB      |
| redis             | 0.50      | 256 MiB      |

**Objective.** End-to-end p99 (payment accepted until its terminal status is recorded at the gateway) under 500 ms, HTTP error rate under 0.1%, no dropped load iterations, and recovery within 120 s after a spike. The recovery check uses end-to-end latency as a stand-in for consumer lag; consumer lag itself was not recorded. The steady and spike payload mix is 80% normal payments, 10% replays of an earlier idempotency key and 10% that fail validation, and 50% of the traffic belongs to one large merchant, which is what makes the record key matter (F-12). The smoke scenario (10 virtual users) spreads traffic evenly instead.

**Scenarios.** The step lists and durations were adjusted to the hardware, as the specification allows, and are recorded with each result:

| Run                                                                        | Scenario parameters                                                     |
|----------------------------------------------------------------------------|-------------------------------------------------------------------------|
| 2026-09-30_baseline_smoke                                                  | 10 virtual users for 60 s                                               |
| 2026-09-30_baseline_spike                                                  | base 50 req/s, burst 500 req/s for 60 s after 30 s, then 120 s recovery |
| 2026-09-30_baseline_spike_recorded (recorded during the dashboard capture) | base 50 req/s, burst 500 req/s for 60 s after 30 s, then 120 s recovery |
| 2026-09-30_baseline_steady                                                 | steps 100, 200, 400, 800 req/s, 60 s each                               |

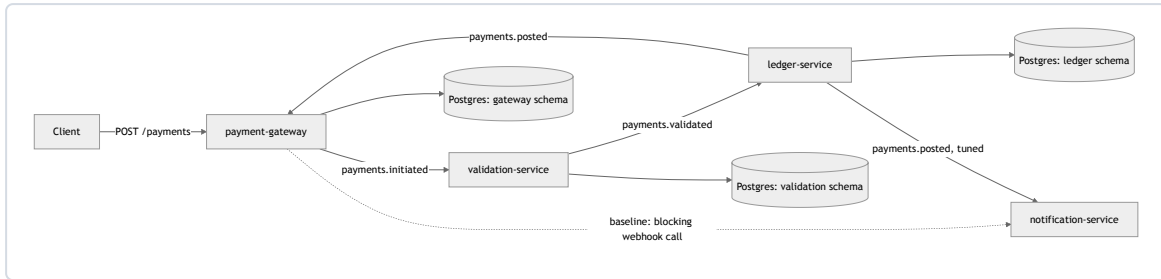
| Run                                                                     | Scenario parameters                                                     |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------|
| 2026-09-30_tuned_smoke                                                  | 10 virtual users for 60 s                                               |
| 2026-09-30_tuned_spike                                                  | base 50 req/s, burst 500 req/s for 60 s after 30 s, then 120 s recovery |
| 2026-09-30_tuned_spike_recorded (recorded during the dashboard capture) | base 50 req/s, burst 500 req/s for 60 s after 30 s, then 120 s recovery |
| 2026-09-30_tuned_steady                                                 | steps 100, 200, 400, 800 req/s, 60 s each                               |
| 2026-09-30_tuned_steady_coldstart-800-3200                              | steps 800, 1600, 2400, 3200 req/s, 60 s each                            |
| 2026-09-30_tuned_steady_extended                                        | steps 400, 800, 1200, 1600 req/s, 60 s each                             |

**Procedure.** Before every run the stack is reset (volumes dropped) and the services are restarted under the profile being measured, so each run starts from the same database and topics. After the load, the run waits for the pipeline to drain, then checks the invariants and captures query plans. End-to-end latencies come from the database ( `updated_at` - `created_at` of the gateway row), not from the services' own counters.

### Run history, stated plainly.

- The baseline was first recorded before the tuned profile existed. Later fixes that also affect baseline behaviour were made (a lock-ordering fix in the ledger, the dead-letter destination, and a read-only validation transaction). The baseline was therefore **re-recorded** at the final revision so that before and after compare identical code. The earlier recording remains in the git history.
- A first steady run was discarded because the machine went to sleep during it; the benchmark script now holds a sleep assertion.
- 2 additional tuned steady runs with higher step lists are kept. In `2026-09-30_tuned_steady_coldstart-800-3200` (first step 800 req/s) and `2026-09-30_tuned_steady_extended` (first step 400 req/s) the first step missed the SLO although the same offered rate met it when reached after lighter steps in another run. That is consistent with warm-up (JIT compilation, connection pools, caches); it was observed once per run and is not proven.
- `2026-09-30_baseline_spike` was updated by `scripts/normalize_results.py`, which moves an unrecovered spike's recovery value from `recovery_seconds_after_burst` to `recovery_at_least_seconds` after the collector was changed to record it that way. The change is recorded in that file under `result_migrations`; no measured value was altered.

### 3. Architecture as found



A client submits a payment with an idempotency key. The gateway records it, assigns a per-debtor sequence number and publishes `payments.initiated`. Validation checks schema, ownership, status, limits and a sanctions stub, and publishes `payments.validated` for every payment, rejections included. The ledger applies payments once (a dedup record written in the same transaction as the postings), in per-debtor sequence order, with double-entry postings and a balance that cannot go negative. It publishes the outcome on `payments.posted`, and the gateway updates the payment status from that event. One Postgres instance holds a schema per service; no service reads another's schema.

**Partitioning as found.** Every topic carries records keyed by merchant identifier, with these partition counts:

| Profile  | payments.initiated | payments.validated | payments.posted | payments.validated.DLT |
|----------|--------------------|--------------------|-----------------|------------------------|
| baseline | 3                  | 3                  | 3               | 3                      |

### 4. Baseline results

**Steady (stepped constant arrival rate).**

| Offered (req/s) | Achieved (req/s) | POST p99 | End-to-end p50 | End-to-end p99 | Dropped iterations | SLO met |
|-----------------|------------------|----------|----------------|----------------|--------------------|---------|
| 100             | 100              | 11 ms    | 6 ms           | 208 ms         | 0                  | yes     |
| 200             | 200              | 6 ms     | 6 ms           | 308 ms         | 0                  | yes     |
| 400             | 400              | 61 ms    | 116.1 s        | 159.2 s        | 0                  | no      |
| 800             | 444              | 5.8 s    | 196.8 s        | 237.4 s        | 21381              | no      |

**Spike.** 50 req/s, then 500 req/s for 60 s, then back. Recovery: not recovered within the observed window (the recovery phase lasted 120 s); the true recovery time is unknown.

**Correctness checks after every baseline run:**

| Run                       | Payments created | All invariants hold | Negative balances | Debits minus credits (minor units) | Non-terminal payments | Sequence conflicts |
|---------------------------|------------------|---------------------|-------------------|------------------------------------|-----------------------|--------------------|
| 2026-09-30_baseline_smoke | 1640             | yes                 | 0                 | 0                                  | 0                     | 0                  |

| Run                                                                        | Payments created | All invariants hold | Negative balances | Debits minus credits (minor units) | Non-terminal payments | Sequence conflicts |
|----------------------------------------------------------------------------|------------------|---------------------|-------------------|------------------------------------|-----------------------|--------------------|
| 2026-09-30_baseline_spike                                                  | 33836            | yes                 | 0                 | 0                                  | 0                     | 0                  |
| 2026-09-30_baseline_spike_recorded (recorded during the dashboard capture) | 33759            | yes                 | 0                 | 0                                  | 0                     | 0                  |
| 2026-09-30_baseline_steady                                                 | 62009            | yes                 | 0                 | 0                                  | 0                     | 0                  |

## 5. Findings

Each finding lists the evidence available, the expected impact, a recommendation, the effort (S, M or L) and the risk of making the change. Impact is **assessed** from the mechanism and the combined measurement; it was not isolated per finding. F-02, F-05, F-09, F-10, F-11 are implemented in the repository but were not part of any recorded benchmark run: the recorded tuned runs carry only the changes listed above, and the recorded baseline runs predate the baseline anti-patterns that go with the others. The text on those findings rests on code, tests and mechanism, and no measured figure in this report is attributed to them.

### F-01 Producer sends synchronously, without batching or compression

- **Evidence.** In the baseline gateway the payment is published with a blocking send inside the database transaction, with `linger.ms` at 0 and no compression ( `PaymentService` , `payment-gateway-baseline.yml` ). The other services rely on Kafka's producer defaults.
- **Impact.** Every request pays a broker round trip and produces small batches. On its own the effect is modest, because F-07 removes the send from the request path.
- **Recommendation.** Publish from an outbox with asynchronous sends and a callback, `linger.ms` around 10, a larger batch size (65536 bytes here) and lz4 compression. Do not linger where a consumer must wait for each acknowledgement before committing its offset (validation).
- **Effort / risk.** S / Low. Partial send failures can reorder events; the ledger orders by sequence number, so this is safe here.

### F-02 Producers allow one request in flight and no idempotence

- **Evidence.** The baseline sets idempotence off and the in-flight request limit to 1, explicitly, in every producing service ( `*-baseline.yml` ); the tuned profile turns idempotence on and allows 5 in flight. Recent clients default to idempotence with several requests in flight, so a real system has to be checked, not assumed. `ProducerConfigTest` reads the effective settings from the running services, and `DuplicateDeliveryTest` injects the same record more than once on every topic to show the consumers absorb it.

- **Impact.** With acknowledgement from all in-sync replicas, a single request in flight per connection serialises produce round trips, so a busy producer is bounded by the round-trip time; without idempotence a retried send can write a duplicate. Assessed from the mechanism, not isolated; it matters most on a replicated cluster, where acknowledgement takes longer.
- **Recommendation.** Enable idempotence and allow several requests in flight (order per partition is kept). It is not end-to-end exactly-once: it only suppresses duplicates within one producer session, so keep the consumer-side deduplication.
- **Effort / risk.** S / Low.

### F-03 One consumer thread regardless of partition count

- **Evidence.** The baseline runs 1 consumer thread per service over 3 partitions ( `*-baseline.yml` ); the tuned profile runs 12 consumer threads. `PartitioningTest` checks the consumer-group membership in both profiles.
- **Impact.** A single thread caps each stage at one record's processing time at a time. Raising concurrency only helps if there are enough partitions (F-12) and enough database connections (F-07).
- **Recommendation.** Match consumer threads to partitions per instance, after sizing the connection pools.
- **Effort / risk.** S / Medium. Real parallelism exposes lock-ordering problems in the ledger; one was found and fixed while doing this work (a drain that took locks out of order).

### F-04 Ledger commits one record per transaction

- **Evidence.** `LedgerService.handle` opens a transaction per record and issues each statement in its own round trip (locks, dedup check, postings, balances, outbox). The tuned `BatchLedgerProcessor` applies a whole poll in one transaction with JDBC batches (up to 250 records per poll here).
- **Impact.** Assessed highest: it multiplies statements and commits per payment on the stage that writes the most rows and takes account locks. This is a judgement from the mechanism; nothing isolates it from the rest.
- **Recommendation.** Batch listener with one ordered lock set per batch, in-memory application against the locked snapshot, dedup inside and across batches, and per-record fallback for poison messages, never dead-lettering a whole batch.
- **Effort / risk.** M / High. This is the most delicate change: it must keep idempotency, sequence ordering and the balance invariant. It is covered by dedicated tests (duplicates, reordering, opposite transfers, a poison record, a database outage).

### F-05 A blocking webhook call inside the status consumer

- **Evidence.** The baseline gateway calls the client's webhook from inside the consumer that records final payment status ( `BlockingWebhookNotifier` ), after the status update has committed. The webhook simulator answers after 5 ms by default, a lab parameter that no recorded run exercised. `NotificationTest` shows a slow webhook delaying later status updates in the baseline but not in the tuned profile, and that a failing webhook never loses or doubles a status update in either.

- **Impact.** With one consumer thread the rate of status updates is capped at one over the webhook's latency, a slow webhook delays every later update, and a run of slow calls can exceed the consumer's poll interval. Notification itself is at-most-once: a crash or a failed call loses it. Assessed high because it can cap the completed-payment rate of the whole pipeline; its share of the combined result was not isolated.
- **Recommendation.** Take the call off the status path: a notification service reads the posted events in its own consumer group, stores them, and delivers with retries, backoff and a dead state, with the receiver deduplicating by payment identifier (at-least-once delivery, effectively once).
- **Effort / risk.** M / Medium. Clients need idempotent receivers, and dead notifications need a repair process.

## F-06 Missing indexes

- **Evidence.** Query plans captured after the steady runs:

| Profile  | Query              | Plan node                                           | Rows removed by filter | Execution time |
|----------|--------------------|-----------------------------------------------------|------------------------|----------------|
| baseline | idempotency lookup | Seq Scan on payments                                | 62008                  | 5.52 ms        |
| baseline | daily outflow      | Seq Scan on postings                                | 110183                 | 5.02 ms        |
| tuned    | idempotency lookup | Index Scan using payments_client_key_uq on payments | none reported          | 0.05 ms        |
| tuned    | daily outflow      | Bitmap Index Scan on postings_account_created_idx   | none reported          | 0.37 ms        |

- **Impact.** The cost of both queries grows with table size, so the baseline degrades as a run (or a day) accumulates rows; the plans above were captured at different table sizes and show the mechanism, not a like-for-like timing.
- **Recommendation.** A unique index on the client and idempotency key, and an index on posting account and creation time. (The payment identifier is already a deterministic primary key; the cost removed is the scan of the key lookup.)
- **Effort / risk.** S / Low.

## F-07 Gateway waits for the broker inside a transaction; pools left at defaults

- **Evidence.** The baseline gateway holds a pooled connection, an advisory lock and the debtor's sequence row lock across the broker round trip, and all pools use the driver default. `OutboxTest` shows the contract difference: with the broker down the baseline answers 503 and stores nothing, while the tuned gateway accepts the payment and publishes it later.
- **Impact.** Under load, request threads queue for connections that are held during network waits.

- **Recommendation.** Write the payment and an outbox row in one short transaction and publish after commit; size the pools for the database (gateway 6, validation 6, ledger 8 here). Those pool sizes are unswept lab choices, not recommendations.
- **Effort / risk.** M / Medium. Clients must accept that a payment can be accepted while the broker is unavailable; load shedding on outbox backlog is recommended and was not implemented.

#### F-08 Lazy per-account limit queries (N+1)

- **Evidence.** The baseline loads each account and then its limits with a separate query, including for the creditor whose limits are unused. `ReferenceDataAccessTest` asserts that the baseline scans the limit table more than once per validation (once for each account) and that the tuned profile does so clearly less than that (a property assertion, not a benchmark figure).
- **Impact.** Extra queries per payment; small on its own once F-13 or a cache is in place.
- **Recommendation.** One projection query joining accounts and their limit, an index on the limit table, and a short-lived in-process cache for the small, rarely changing rate table.
- **Effort / risk.** S / Low.

#### F-09 A new JSON mapper per message and full payloads logged at info level

- **Evidence.** The baseline builds a JSON mapper for every message and logs every event at info level with its full payload, in every service. `SerializationLoggingTest` counts the mappers built and captures the log events in both profiles.
- **Impact.** Building a mapper costs far more than using one, and log volume grows with every message. Full payloads also put account identifiers and amounts into the logs, a data-protection problem whatever the speed. The speed effect is assessed as modest and was not isolated.
- **Recommendation.** One shared, configured mapper; log identifiers at debug level and keep info for what an operator needs to see.
- **Effort / risk.** S / Low.

#### F-10 Platform threads for blocking calls and default JVM sizing

- **Evidence.** The baseline gateway serves requests on platform threads that block on the database and the broker, and every JVM runs with its defaults. The tuned profile serves requests on virtual threads ( `ThreadingTest` counts requests by thread kind) and starts the JVMs with `-Xms512m -Xmx512m -XX:+UseG1GC -XX:MaxGCPauseMillis=50`. The services run on the host, not in containers, so this is a fixed heap rather than a container-aware percentage. The collectors compared on the same steady ladder:

*(no recorded run pairs G1 with ZGC: the change was implemented after the recorded benchmark runs, so this comparison has no data)*

- **Impact.** Virtual threads make blocking calls cheap in the gateway, and a fixed heap avoids resizing pauses. Assessed as moderate and not isolated; the collector comparison, once recorded, is a single pair of runs.

- **Recommendation.** Virtual threads for blocking request handling (watch for pinning under synchronized code), a heap sized from observed use, and a collector chosen from measurements rather than habit.
- **Effort / risk.** S / Medium: pinning and native memory behaviour must be checked under the real workload.

## F-11 Every posting locks one settlement account

- **Evidence.** Every payment is posted through a settlement account (debtor to settlement to creditor). The baseline uses one settlement account for all payments (`SettlementAccounts`); the tuned profile picks one of 16 shard accounts per transaction from the source partition. `SettlementTest` shows every settlement account back at nothing after every commit, a fixed number of legs per posted payment, and the rejections for reserved and missing settlement accounts. The tuned profile against the tuned profile with only this change switched off (an ablation):

*(no recorded run pairs Tuned with Tuned without F-11: the change was implemented after the recorded benchmark runs, so this comparison has no data)*

- **Impact.** One account row locked by every payment serialises them as soon as more than one consumer thread runs, so the baseline (one consumer thread) cannot show it and the tuned profile can. Assessed high once concurrency is raised; the ablation table above, once it has data, is the evidence for its size.
- **Recommendation.** Shard hot transit accounts and choose the shard per transaction, not per payment, so a batch does not lock every shard. Add the shard to the same ordered lock set as the other accounts so no new deadlock is possible, and reconcile periodically.
- **Trade-off.** The aggregate position must be read across shards. Because the settlement legs net out inside each transaction there is nothing to sweep; a reconciliation job records the position and warns if any shard is not at nothing.
- **Effort / risk.** M / Medium.

## F-12 Records keyed by merchant on few partitions

- **Evidence.** Baseline keys every record by merchant on the partition counts in section 3; the tuned profile keys by debtor account, with the counts of both profiles below. `PartitioningTest` checks keys and counts in both profiles. The steady and spike load sends 50% of its traffic through one merchant.

| Profile  | payments.initiated | payments.validated | payments.posted | payments.validated.DLT |
|----------|--------------------|--------------------|-----------------|------------------------|
| baseline | 3                  | 3                  | 3               | 3                      |
| tuned    | 12                 | 12                 | 12              | 12                     |

- **Impact.** A large merchant concentrates work on one partition and therefore one consumer thread; too few partitions also cap useful concurrency (F-03). Whether a hot partition limited the baseline was not measured (no per-partition lag was captured).

- **Recommendation.** Key by debtor account and size partitions from measured per-partition throughput against the target rate; partitions can be added later but never removed, so plan the migration. Per-debtor ordering does not depend on the key (sequence numbers), which is what makes the change safe.
- **Effort / risk.** M / Medium. A single very busy debtor becomes a hot partition.

### F-13 Account data read from Postgres on every validation

- **Evidence.** The baseline reads status, limits and ownership from Postgres for every payment. The tuned profile serves them from a Redis cache with versioned invalidation events, a time-to-live of 60 s as a backstop, single-flight loading and fail-open behaviour; `CacheTest` covers invalidation, the stale-reader race, stampede protection and a Redis outage. The cache hit ratio was **not** captured in the benchmark runs.
- **Impact.** Assessed moderate and only after F-06 and F-08 have removed the cheap wins; there is no measurement of its share of the combined result.
- **Recommendation.** Defer until the effect of the cheaper changes is measured and a hit ratio can be observed. The trade-off is staleness: a blocked account can still be approved until its cached copy is invalidated or expires. Any adoption needs an explicit staleness tolerance from the business.
- **Effort / risk.** L / High.

## 6. Architecture observations

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These are separate from the performance findings: they concern how the system is put together, and would matter even at low load.

- **Service boundaries.** The split into gateway, validation and ledger follows ownership of data well: the gateway owns payment status, validation owns reference data, the ledger owns balances and postings, with one Postgres schema each. Account attributes (currency, owner) exist in both the validation and the ledger schemas with no synchronisation mechanism; in the lab both are seeded, in production they would drift.
- **Synchronous coupling.** In the baseline the availability of `POST /payments` depends on the broker, because the gateway publishes inside its request. The outbox removes that dependency at the cost of accepting payments while the pipeline is stalled; the client contract and the alerting need to reflect that.
- **Retry and dead-letter strategy.** The validation and ledger consumers retry and then dead-letter a failing record. A dead-lettered payment leaves a gap in its debtor's sequence, and every later payment for that debtor waits behind the gap. There is no re-drive tooling and nothing consumes or alerts on the dead-letter topics. The tuned batch consumer retries transient database errors without limit and only dead-letters a record that fails on its own. Recommendation: monitor the dead-letter topics, provide a re-drive path, and define how a sequence gap is repaired.
- **Unhandled status updates.** The gateway consumer that records final payment status has no dead-letter handling and there is no dead-letter topic for `payments.posted`. With the framework's default handler a failing status update is retried briefly without delay and then

logged and skipped, which would leave the payment shown as accepted indefinitely with no trace except a log line. Recommendation: give this consumer the same retry, dead-letter and alerting treatment as the others.

- **Schema evolution.** Events carry a schema version and consumers ignore unknown fields, which tolerates additive change. There is no schema registry and no compatibility check, so a renamed or removed field breaks consumers at runtime. Recommendation: a registry or consumer-driven contract tests before the first breaking change.
- **Delivery semantics.** Delivery is at-least-once everywhere; correctness comes from deterministic identifiers, the ledger's dedup record and monotonic status updates. This is sound and well tested, but it means the service counters count attempts, so they must not be used for business reporting (ADR-0003).
- **Notification delivery.** In the baseline the gateway calls the client's webhook itself, after the status update has committed. Delivery is therefore at-most-once (a crash or a failed call loses the notification) and a slow webhook delays every later status update on the same consumer thread. The tuned notification service stores each posted event, delivers it with retries and backoff, and relies on the receiver deduplicating by payment identifier, so delivery is at-least-once and effectively once. Webhooks are not ordered; there is one terminal event per payment, so that is harmless.
- **Tenancy and access.** The client identity is a trusted request header, and the tuned profile adds a token-protected administration endpoint on the service port. Both are lab conveniences, not designs to copy.
- **Availability.** The recorded runs used a single Kafka broker with replication factor 1 and a single database instance; the database is a lab limit and its failover was not tested. Broker-failure behaviour was not tested in the recorded runs.

## 7. Prioritised remediation roadmap

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Impact is assessed on a scale of 1 to 5 and effort is S, M or L; quadrants use an impact of 4 or more as high and effort S as small. Where the score and the recommended order disagree (F-03 scores as a quick win but depends on F-12, F-07), the sequenced plan follows the dependency.



| ID   | Finding                                                                                    | Area                     | Assessed impact | Effort | Risk   | Quadrant      |
|------|--------------------------------------------------------------------------------------------|--------------------------|-----------------|--------|--------|---------------|
| F-04 | Ledger commits one record per transaction with one statement per round trip                | Ledger writes            | 5 of 5          | M      | High   | Major project |
| F-06 | No indexes on the idempotency lookup and on the daily-outflow query                        | Database                 | 4 of 5          | S      | Low    | Quick win     |
| F-03 | One consumer thread per service regardless of partition count                              | Kafka consumer           | 4 of 5          | S      | Medium | Quick win     |
| F-05 | A blocking webhook call sits inside the status consumer                                    | Consumer and integration | 4 of 5          | M      | Medium | Major project |
| F-07 | Gateway holds a connection and locks while it waits for the broker; pools left at defaults | Gateway and pools        | 4 of 5          | M      | Medium | Major project |
| F-11 | Every posting locks one settlement account                                                 | Ledger contention        | 4 of 5          | M      | Medium | Major project |
| F-12 | Records keyed by merchant on few partitions                                                | Topic design             | 4 of 5          | M      | Medium | Major project |

| ID   | Finding                                                                             | Area                      | Assessed impact | Effort | Risk   | Quadrant       |
|------|-------------------------------------------------------------------------------------|---------------------------|-----------------|--------|--------|----------------|
| F-10 | Platform threads for blocking calls and default JVM sizing                          | JVM                       | 3 of 5          | S      | Medium | Fill-in        |
| F-13 | Every validation reads account status, limits and rates from Postgres               | Caching                   | 3 of 5          | L      | High   | Defer or avoid |
| F-01 | Producer sends one record at a time, synchronously, without batching or compression | Kafka producer            | 2 of 5          | S      | Low    | Fill-in        |
| F-02 | Producers allow one request in flight and no idempotence                            | Kafka producer            | 2 of 5          | S      | Low    | Fill-in        |
| F-08 | Validation loads limits with lazy per-account queries (N+1)                         | Validation reads          | 2 of 5          | S      | Low    | Fill-in        |
| F-09 | A new JSON mapper per message and full payloads logged at info level                | Serialization and logging | 2 of 5          | S      | Low    | Fill-in        |

**Sequenced plan.** The order respects dependencies within and across phases: partitions and pools before consumer threads, indexes before the outbox that relies on them, the outbox before asynchronous publishing, the projection query before the cache.

| Sequence    | ID   | Change                                                                                     | Effort | Risk   | Depends on |
|-------------|------|--------------------------------------------------------------------------------------------|--------|--------|------------|
| Quick wins  | F-02 | Producers allow one request in flight and no idempotence                                   | S      | Low    | none       |
| Quick wins  | F-06 | No indexes on the idempotency lookup and on the daily-outflow query                        | S      | Low    | none       |
| Quick wins  | F-08 | Validation loads limits with lazy per-account queries (N+1)                                | S      | Low    | none       |
| Quick wins  | F-09 | A new JSON mapper per message and full payloads logged at info level                       | S      | Low    | none       |
| Next sprint | F-05 | A blocking webhook call sits inside the status consumer                                    | M      | Medium | none       |
| Next sprint | F-07 | Gateway holds a connection and locks while it waits for the broker; pools left at defaults | M      | Medium | F-06       |
| Next sprint | F-10 | Platform threads for blocking calls and default JVM sizing                                 | S      | Medium | none       |
| Next sprint | F-12 | Records keyed by merchant on few partitions                                                | M      | Medium | none       |
| Next sprint | F-01 | Producer sends one record at a time, synchronously, without batching or compression        | S      | Low    | F-07       |
| Next sprint | F-03 | One consumer thread per service regardless of partition count                              | S      | Medium | F-12, F-07 |

| Sequence    | ID   | Change                                                                      | Effort | Risk   | Depends on |
|-------------|------|-----------------------------------------------------------------------------|--------|--------|------------|
| Next sprint | F-11 | Every posting locks one settlement account                                  | M      | Medium | F-03       |
| Structural  | F-04 | Ledger commits one record per transaction with one statement per round trip | M      | High   | none       |
| Structural  | F-13 | Every validation reads account status, limits and rates from Postgres       | L      | High   | F-08       |

## 8. Tuned results against baseline

**Configuration differences** (the full tuned configuration is in Appendix A):

| Setting                                       | Baseline                                     | Tuned                                                            | Finding    |
|-----------------------------------------------|----------------------------------------------|------------------------------------------------------------------|------------|
| Record key                                    | merchant identifier                          | debtor account identifier                                        | F-12       |
| Partitions per topic                          | 3                                            | 12                                                               | F-12       |
| Consumer threads per service                  | 1                                            | 12                                                               | F-03       |
| Producer linger (gateway)                     | 0 ms                                         | 10 ms                                                            | F-01       |
| Producer batch size / compression             | Kafka default / none                         | 65536 bytes / lz4                                                | F-01       |
| Gateway publishing                            | blocking send inside the request transaction | outbox row, asynchronous publisher                               | F-07, F-01 |
| Database pool (gateway / validation / ledger) | driver default                               | 6 / 6 / 8                                                        | F-07       |
| Ledger processing                             | one record per transaction                   | batches of up to 250 records, JDBC batch writes                  | F-04       |
| Indexes                                       | primary keys only                            | idempotency key, postings by account and time, limits by account | F-06, F-08 |
| Limits lookup                                 | lazy query per account                       | one projection query, cached rates                               | F-08       |
| Account data source                           | Postgres on every validation                 | Redis cache, 60 s time-to-live, event invalidation               | F-13       |

**Steady.** The same offered rates against both profiles:

| Offered (req/s) | Baseline e2e p50 | Baseline e2e p99 | Baseline SLO | Tuned e2e p50 | Tuned e2e p99 | Tuned SLO |
|-----------------|------------------|------------------|--------------|---------------|---------------|-----------|
| 100             | 6 ms             | 208 ms           | yes          | 112 ms        | 162 ms        | yes       |
| 200             | 6 ms             | 308 ms           | yes          | 112 ms        | 158 ms        | yes       |
| 400             | 116.1 s          | 159.2 s          | no           | 100 ms        | 157 ms        | yes       |
| 800             | 196.8 s          | 237.4 s          | no           | 104 ms        | 174 ms        | yes       |

## Spike.

| Phase   | Offered (req/s) | Baseline e2e p50 | Baseline e2e p99 | Baseline dropped | Tuned e2e p50 | Tuned e2e p99 | Tuned dropped |
|---------|-----------------|------------------|------------------|------------------|---------------|---------------|---------------|
| warm    | 50              | 8 ms             | 125 ms           | 0                | 112 ms        | 335 ms        | 0             |
| burst   | 500             | 64.0 s           | 106.8 s          | 0                | 108 ms        | 167 ms        | 0             |
| recover | 50              | 60.1 s           | 106.8 s          | 0                | 109 ms        | 152 ms        | 0             |

**Stage observations** (peaks over the whole steady run, which includes the overloaded steps, so they show where pressure appeared, not what limited each step):

*(no stage metrics are recorded for these runs)*

**Smoke** (10 virtual users, correctness check):

| Profile  | Payments created | End-to-end p50 | p95    | p99    | HTTP error rate | p99 under SLO |
|----------|------------------|----------------|--------|--------|-----------------|---------------|
| baseline | 1640             | 47 ms          | 76 ms  | 125 ms | 0.00%           | yes           |
| tuned    | 1660             | 99 ms          | 139 ms | 171 ms | 0.00%           | yes           |

**Correctness checks after every tuned run:**

| Run                                                                     | Payments created | All invariants hold | Negative balances | Debits minus credits (minor units) | Non-terminal payments | Sequence conflicts |
|-------------------------------------------------------------------------|------------------|---------------------|-------------------|------------------------------------|-----------------------|--------------------|
| 2026-09-30_tuned_smoke                                                  | 1660             | yes                 | 0                 | 0                                  | 0                     | 0                  |
| 2026-09-30_tuned_spike                                                  | 33808            | yes                 | 0                 | 0                                  | 0                     | 0                  |
| 2026-09-30_tuned_spike_recorded (recorded during the dashboard capture) | 33706            | yes                 | 0                 | 0                                  | 0                     | 0                  |
| 2026-09-30_tuned_steady                                                 | 81001            | yes                 | 0                 | 0                                  | 0                     | 0                  |
| 2026-09-30_tuned_steady_coldstart-800-3200                              | 238631           | yes                 | 0                 | 0                                  | 0                     | 0                  |
| 2026-09-30_tuned_steady_extended                                        | 209731           | yes                 | 0                 | 0                                  | 0                     | 0                  |

## Reading these tables.

- The tuned profile has a higher median end-to-end latency than the baseline at low load (the smoke run and the lowest steady step). The likely causes are the outbox polling on the gateway and the ledger and producer lingering, but that was not verified. It is a real

trade-off: a higher latency floor at low load in exchange for capacity at high load.

- Steps that missed the objective in the baseline column are dominated by queueing: those numbers describe an overloaded pipeline, not its service time. Baseline steps are also not independent, because its cost grows as the tables fill during a run.
- Additional tuned runs with higher step lists, kept as recorded:

| Run                                        | First step | Offered (req/s) | Achieved (req/s) | POST p99 | End-to-end p50 | End-to-end p99 | Dropped iterations | SL me |
|--------------------------------------------|------------|-----------------|------------------|----------|----------------|----------------|--------------------|-------|
| 2026-09-30_tuned_steady_coldstart-800-3200 | yes        | 800             | 795              | 586 ms   | 118 ms         | 4.1 s          | 283                | no    |
| 2026-09-30_tuned_steady_coldstart-800-3200 | no         | 1600            | 1339             | 2.3 s    | 145.2 s        | 169.3 s        | 15655              | no    |
| 2026-09-30_tuned_steady_coldstart-800-3200 | no         | 2400            | 1163             | 3.0 s    | 102.7 s        | 128.7 s        | 74207              | no    |
| 2026-09-30_tuned_steady_coldstart-800-3200 | no         | 3200            | 1112             | 4.5 s    | 49.8 s         | 76.7 s         | 125289             | no    |
| 2026-09-30_tuned_steady_extended           | yes        | 400             | 400              | 212 ms   | 104 ms         | 1.6 s          | 9                  | no    |
| 2026-09-30_tuned_steady_extended           | no         | 800             | 800              | 53 ms    | 126 ms         | 311 ms         | 0                  | yes   |
| 2026-09-30_tuned_steady_extended           | no         | 1200            | 1200             | 126 ms   | 835 ms         | 33.2 s         | 0                  | no    |
| 2026-09-30_tuned_steady_extended           | no         | 1600            | 1481             | 1.7 s    | 34.1 s         | 59.3 s         | 7120               | no    |

**Gatling** (the same workload driven by Gatling instead of k6, to show the results do not depend on the load tool):

*(no Gatling runs are recorded)*

**Soak** (a long steady rate; drift would show as a rising p99 across windows or as growth in heap, threads or pool use, but a single soak run is an observation, not proof of the absence of leaks):

*(no soak runs are recorded)*

**Profiling** (Java Flight Recorder samples running threads on a fixed interval: the CPU view shows where Java code ran, the native view where threads were blocked in native calls such as socket reads. It shows where sampled time went, not everything a thread waited on):

*(no profiling runs are recorded)*

## Broker failure tests

*(no broker-failure runs are recorded)*

## 9. Limits of this evidence

---

- One machine, most data points from a single run, load generator and services sharing the host. Differences at the level of a step are meaningful; small differences are not.
- The limits are known only to step resolution, so any ratio is a range. The offered rate at which dropped iterations appear reflects latency backing up into the load generator; host contention may contribute and was not isolated.
- The combination of the 8 changes (F-01, F-03, F-04, F-06, F-07, F-08, F-12, F-13) in the recorded tuned runs was measured. The effect of each change alone was not isolated. The per-finding impact in section 7 is an informed assessment. An ablation (tuned with one change reverted) is the way to measure each change on its own and is recommended before investing in the larger items.
- No consumer lag, CPU, garbage-collection or lock-wait data was captured for these runs, so which stage limits either profile is not established. The dashboard exists ( `grafana/dashboards` ) but no snapshots of it were stored with the results.
- An earlier recording of the baseline spike scenario (commit `eeeb939` ), made before the baseline-affecting fixes, recovered within the window; the standard recording used in the tables above did not. A further recording, taken while the dashboard was being captured ( `2026-09-30_baseline_spike_recorded` ), also recovered. The recordings disagree, which shows run-to-run variance in this scenario for the baseline; no cause was attributed and no single recovery time is claimed.
- The query plans were captured after the runs at different table sizes.
- Replays of an earlier idempotency key create no payment, so payments created are about 90% of the requests sent in the steady and spike runs.

## Appendix A. Tuned configuration (as recorded with the run; comment lines removed by the recorder)

---

**payment-gateway** ( `payment-gateway-tuned.yml` , comments removed by the recorder)

```
lab:
  topics:
    partitions: 12
  tuning:
    complete: true  # every MVP finding (F-01, F-03, F-04, F-06, F-07, F-
                     08, F-12, F-13) is implemented; run-benchmark.sh checks this
    f01: true
    f07: true
    f12: true
spring:
  kafka:
    listener:
      concurrency: 12
    producer:
      compression-type: lz4
      batch-size: 65536
      properties:
        linger.ms: 10
  datasource:
    hikari:
      maximum-pool-size: 6
      connection-timeout: 2000
  flyway:
    locations: classpath:db/gateway,classpath:db/gateway-tuned
```

**validation-service** ( validation-service-tuned.yml , comments removed by the recorder)

```

lab:
  admin:
    token: ${LAB_ADMIN_TOKEN:lab-admin-local-only} # F-13 admin endpoint
      (lab only); placeholder value
  cache:
    namespace: v1 # F-13: bump to drop every cached entry (key
      versioning)
    ttl-seconds: 60 # F-13: backstop if an invalidation event is missed
  topics:
    partitions: 12
  tuning:
    f08: true
    f12: true
    f13: true
  spring:
    data:
      redis:
        host: ${REDIS_HOST:localhost}
        port: ${REDIS_PORT:6390}
        timeout: 100ms # F-13: fail open quickly when Redis is slow
          or down
        connect-timeout: 100ms
      kafka:
        listener:
          concurrency: 12
        producer:
          compression-type: lz4
          batch-size: 65536
          properties:
            linger.ms: 0
      datasource:
        hikari:
          maximum-pool-size: 6
          connection-timeout: 5000
      flyway:
        locations: classpath:db/validation,classpath:db/validation-tuned
  management:
    health:
      redis:
        enabled: true

```

**ledger-service** ( `ledger-service-tuned.yml`, comments removed by the recorder)

```

lab:
  topics:
    partitions: 12
  outbox:
    poll-ms: 50      # F-04: the outbox flush no longer runs on (and
                     serialises) the listener threads
  tuning:
    f01: true
    f04: true
    f12: true
  spring:
    kafka:
      listener:
        type: batch    # F-04
        concurrency: 12
      consumer:
        max-poll-records: 250      # F-04: batch size per poll
        properties:
          max.poll.interval.ms: 600000 # F-04: a batch may retry a database
                                         outage for a long time
      producer:
        compression-type: lz4
        batch-size: 65536
        properties:
          linger.ms: 10
    datasource:
      hikari:
        maximum-pool-size: 8
        connection-timeout: 60000
        data-source-properties:
          rewriteBatchedInserts: true # F-04: multi-row INSERT rewriting
                                       for JDBC batches
    flyway:
      locations: classpath:db/ledger,classpath:db/ledger-tuned

```

## Appendix B. Raw data

| Run                                                                        | Git sha | Files                                                                                                              |
|----------------------------------------------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------|
| 2026-09-30_baseline_smoke                                                  | 06ce6f4 | <a href="#">result.json</a> · <a href="#">summary.json</a> · <a href="#">env.txt</a> · <a href="#">explain.txt</a> |
| 2026-09-30_baseline_spike                                                  | 06ce6f4 | <a href="#">result.json</a> · <a href="#">summary.json</a> · <a href="#">env.txt</a> · <a href="#">explain.txt</a> |
| 2026-09-30_baseline_spike_recorded (recorded during the dashboard capture) | 8862be4 | <a href="#">result.json</a> · <a href="#">summary.json</a> · <a href="#">env.txt</a> · <a href="#">explain.txt</a> |
| 2026-09-30_baseline_steady                                                 | 06ce6f4 | <a href="#">result.json</a> · <a href="#">summary.json</a> · <a href="#">env.txt</a> · <a href="#">explain.txt</a> |

| Run                                                                     | Git sha | Files                                                                                                              |
|-------------------------------------------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------|
| 2026-09-30_tuned_smoke                                                  | 06ce6f4 | <a href="#">result.json</a> · <a href="#">summary.json</a> · <a href="#">env.txt</a> · <a href="#">explain.txt</a> |
| 2026-09-30_tuned_spike                                                  | 06ce6f4 | <a href="#">result.json</a> · <a href="#">summary.json</a> · <a href="#">env.txt</a> · <a href="#">explain.txt</a> |
| 2026-09-30_tuned_spike_recorded (recorded during the dashboard capture) | 8862be4 | <a href="#">result.json</a> · <a href="#">summary.json</a> · <a href="#">env.txt</a> · <a href="#">explain.txt</a> |
| 2026-09-30_tuned_steady                                                 | 06ce6f4 | <a href="#">result.json</a> · <a href="#">summary.json</a> · <a href="#">env.txt</a> · <a href="#">explain.txt</a> |
| 2026-09-30_tuned_steady_coldstart-800-3200                              | 958b37b | <a href="#">result.json</a> · <a href="#">summary.json</a> · <a href="#">env.txt</a> · <a href="#">explain.txt</a> |
| 2026-09-30_tuned_steady_extended                                        | 958b37b | <a href="#">result.json</a> · <a href="#">summary.json</a> · <a href="#">env.txt</a> · <a href="#">explain.txt</a> |

## Appendix C. How this report was produced

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`scripts/run-benchmark.sh` produced each result folder; `scripts/collect_results.py` derived `result.json`; `scripts/generate_report.py` filled this document from those files and the service configuration. `scripts/lint_report_numbers.py` rejects any number or number word in the template prose (and in the findings list) that is not a generated placeholder, and CI fails if this file differs from a fresh generation.